

Article

Water–Energy–Milk Nexus: Empirical Evidence from Saudi Arabia

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Abstract: Dairy farming plays a crucial role in Saudi Arabia’s agricultural industry. However, the intensive milk production process exerts pressure on local water and energy resources. This study aims to examine the impact of water stress and renewable energy consumption shocks on milk production in Saudi Arabia by using data from 2000 to 2021. The empirical analysis used the VAR model, Granger causality, forecast error variance decompositions (FEVDs), and impulse response functions (IRFs). The presence of a negative significant interdependence between total milk production and water stress levels in agriculture was observed. Significant bidirectional causality relationships among the variables were noted. The FEVD results show that water stress levels in agriculture are becoming a more dominant driver of variations in total milk production in Saudi Arabia, while the empirical evidence of the IRFs implies that milk production increases when both water stress levels and renewable energy are present. The adoption of water recycling and reuse systems on dairy farms can help farmers to improve water use efficiency. The encouragement of decision makers to formulate policies to support sustainable water resource management, reduce environmental impact, accelerate technological advancements, and initiate positive socioeconomic outcomes for the dairy industry is highly recommended.

Keywords: milk production; water stress; water resource management; VAR; Saudi Arabia



Citation: Elzaki, R.M.; Al-Mahish, M.; Alzahrani, F. Water–Energy–Milk Nexus: Empirical Evidence from Saudi Arabia. *Water* **2024**, *16*, 1538. <https://doi.org/10.3390/w16111538>

Academic Editors: Didier Orange and Yizi Shang

Received: 17 February 2024

Revised: 15 May 2024

Accepted: 23 May 2024

Published: 27 May 2024



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1. Introduction

Water scarcity is increasing due to the local consequences of physical water stress, combined with the rapid expansion and broadcasting of freshwater pollution [1]. From 2010 until now, around 50% of the world’s population has been living under water stress at least one month per year, and scarcity will greatly worsen in regions where water is already in short supply, such as the Middle East and the Sahel in Africa [2,3]. The Food and Agriculture Organization of the United Nations (FAO) reported that by 2025, 1800 million people are predicted to live with absolute water scarcity and two-thirds of the world population could be under water stress [4]. The increasing population will contribute to escalating the scarcity of water, energy, and food [5]; hence, the global population continues to rise, and there is a rapid increase in water and energy demand to fulfill the increasing demand for food [6,7].

The connection among milk production, water stress, and renewable energy is multidimensional and highlights the future of maintainable practices in the dairy industry. In the context of milk production, incorporating renewable energy sources on dairy farms can help reduce the dependence on fossil fuel energy, which often requires significant water resources for extraction and cooling processes. This shift to renewable resources can help alleviate water stress associated with conventional energy production [8,9].

Dairy farming is of great significance to Saudi Arabia and constitutes a vital sector within the country’s agricultural industry. The total milk production in 2021 reached

34.27 thousand tons with an improvement percentage of 2.80% from 2019 (31.52 thousand tons). The milk production industry in Saudi Arabia can continue to meet domestic demand, reduce import dependency, increase milk exports, and contribute to the country's food security goals by promoting the country's vision (2030) by reducing ecological impacts.

From Figure 1, it seems that the dairy sector has seen steady growth in Saudi Arabia, and the peak of milk production from camel, cattle, and goat types was observed in 2021. The positive trend of milk production in recent years may be related to the increasing demand for milk and the adoption of modern farming systems, which has likely encouraged dairy farmers to expand their operations and increase production. Also, we note from Figure 1 that sheep milk reached its maximum in 2014; this could be due to specific market conditions or changes in farming practices during this period. Even though Saudi Arabia has attained self-sufficiency in milk production [10], there are still several challenges and problems faced by this vital industry.

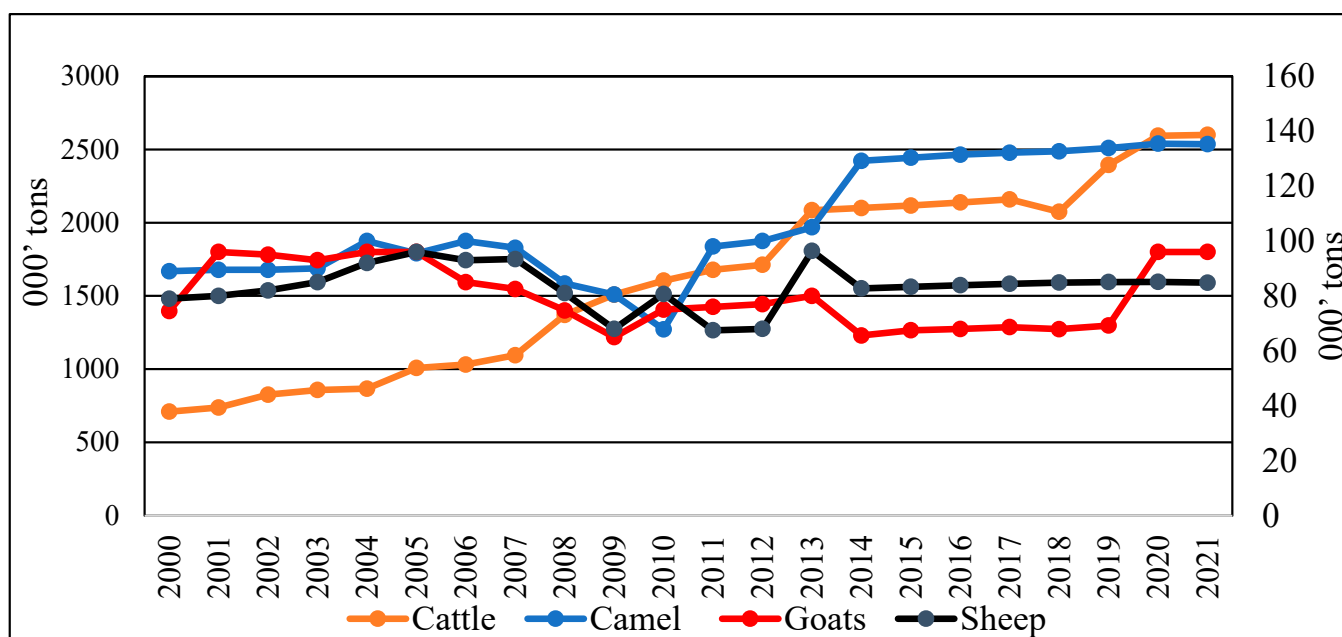


Figure 1. Raw milk production in Saudi Arabia (000' tons). Note: (1) The right axis represents goat and sheep milk production, and (2) the left axis represents camel and cattle milk.

In recent decades, Saudi Arabia has faced challenges regarding inadequate rainfall, prolonged droughts, and fluctuating climate, which affect water stress levels due to insufficient water supply [11]. Those challenges will threaten agricultural production. For instance, milk production requires significant amounts of water for various purposes, including water consumption for livestock rearing and other purposes [12,13]. This can conflict with other water users for other purposes, such as agricultural activities, services, and industries that rely on the same water sources. In Saudi Arabia, where water resources are limited [14], water stress can be a crucial factor affecting agricultural activities, including dairy farming. If water stress increases over time, it indicates a worsening situation where the demand for water in agriculture, including livestock and dairy production, is outpacing the available water supply.

Milk production requires energy for various processes, including milking, fences' cooling, processing, and transportation [15,16]. Nonrenewable energy, such as fossil fuels, is commonly used in these activities, resulting in increasing emissions and contributing to climate change. The milk production process itself does not directly rely on renewable energy sources, and dairy farms may require financial support or incentives to invest in renewable energy infrastructure [17]. By considering these problems and challenges, this study aims to investigate the dynamic relationship among total milk production, water

stress level, and renewable energy in Saudi Arabia. Likewise, this study aims to analyze the impact of the shocks of water stress and renewable energy on the actual milk production in Saudi Arabia.

The current paper will contribute to the development and promotion of the sustainable dairy sector in several ways. Firstly, by examining the complex relationships between milk production, water stress, and renewable energy consumption, policymakers can improve regulations, incentives, and support mechanisms that boost resource-efficient practices. Secondly, researchers and stakeholders can make an effort towards more resilient, resource-efficient, and environmentally friendly methods of milk production. Thirdly, this study can support policy decision making for promoting renewable energy adoption, water conservation, and sustainable farming practices. This helps ensure the long-term feasibility of the dairy industry sector while reducing its ecological footprint. Finally, the outcomes from the empirical results will enrich the literature about the acknowledgment and information of the connection among milk production, water stress, and renewable energy.

This paper is organized as follows: Following the introduction in Section 1, a literature review is briefly discussed in Section 2, Section 3 explores the data and methods applied, Section 4 explains the results of the estimated models, and finally, conclusions and policy implications are presented in Section 5.

2. Literature Review

The hypothetical predictions about the connection among milk production, renewable energy, and water stress can vary depending on several factors, including topographical location, climate change, land use changes, technological adoption, and resource accessibility [18–20].

2.1. Milk–Water Nexus

There are many established studies examining the relationship between water and milk/dairy production. In general, studies can be categorized into four broad research areas: (1) studies that estimated and compared the water footprint of milk yield in diverse countries or production systems, (2) studies that examined the impact of drinking water access/intake on milk production, (3) studies that examined the influence of drinking water quality on milk production, and (4) studies that looked at the impacts of climate change or farmers' perception about climate change on milk production.

The first category is the most common type of study available on the milk–water nexus. For example, ref. [21] compared water use in milk production and intensive, grazing, and small-scale milk production systems in various countries by applying the Technology Impact Policy Impact Calculation (TIPIC) based on the Farm Level Impact of Policy Simulations Model (FLIPSIM). They found that rainwater was the dominant type of water used in milk production across the countries, with African small-scale farms being the largest users of rainwater. In addition, intensive feedlot farms in the Middle East were the largest users of surface and groundwater. Other studies used similar methods of analysis and obtained the same results [22–27].

A few studies also examined the impact of drinking water access/intake on milk production [28]. Recent studies [29,30] examined the impact of providing unrestricted water during the grazing period and found that providing unrestricted access to drinking water increases daily milk production by 5% to 10%. Studies also examined drinking water quality's impact on milk production through the reduction in water intake. The authors of [31,32] performed a review on the effects of water quality on livestock productivity and their performance and confirmed that the quality of drinking water has a significant impact on milk production and livestock health.

The last category of studies examines the impacts of climate change and/or farmers' perception of climate change on milk production. Most of these studies focused on the impact of weather factors on dairy production [33,34] and considered the humidity index, temperature, droughts, and participation. For example, Abbas et al. [35] conducted a

study in Pakistan by using an Ordered Probit model applying the discrete choice type; they indicated that farmers confirmed that drought is one of the major climatic risks that severely affect all aspects of dairy production and showed that climate change had significant influences on milk quantity. Furthermore, most of the studies investigated the influence of water stress on crop production and/or crop growth [36,37]. Crop production is negatively and significantly affected by drought conditions [38].

2.2. Milk–Energy Nexus

In the context of the milk–energy connection, a few scholars examined the association between energy and milk production. Ref. [39] examined energy efficiency in milk production for dairy enterprises in Turkey, applying a stratified random sampling method by examining the linear programming (LP) method. They found that less than 10% of total energy input (e.g., labor, machine, diesel fuel, etc.) per cow was direct and more than 90% was indirect energy. They indicated that most of the total energy output (milk and fertilizer) was from milk production. They also calculated energy productivity and efficiency and found them to be higher and lower than those in other studies, respectively. Furthermore, ref. [40] reviewed studies related to energy consumption on dairy farms, where on average, more than 20% of electricity was used in milking and milk cooling systems. Another strand of the literature is focused on the use or potential use of renewable energy on dairy farms. We observed that most of the studies were involved in the carbon footprint issue linked to the use of renewable energy systems in milk production and confirmed a decline in carbon footprint as a result of using renewable energy sources in milk production or industry [18,19,40,41].

2.3. Milk–Water–Energy Nexus

A large number of studies have examined the environmental impacts of dairy production, using the environmental life cycle assessment method to estimate the resources used, including energy and water, in the production process [42].

A recent study created a water–energy–food (WEF) nexus index by using a cluster ranking to identify the farms with the worst WEF nexus indexes for ranking dairy farms based on carbon emissions, water demand, energy requirements, and milk production [43]. By using this index, the authors suggested a plan of action based on the farm ranking. Ref. [44] analyzed the first- and second-order simple and partial correlation coefficients to examine the economic relationship among energy, water use, and plant and animal food production (including milk) in Saudi Arabia. They found that changes in predicted water, power, or fuel consumption resulted in changes in the index of plant and animal food production.

We concluded from the existing literature that significant limitations persist in the existing literature, since most studies examined the connection among milk production, water, and energy while neglecting the impacts of renewable energy and water stress on milk production. The novelty of this study differs from previous studies in terms of the methodology applied. While most previous studies have mostly focused on descriptive and simple econometric analyses, our study implements multivariate approaches. Specifically, we applied the vector autoregressive (VAR) multivariate model, which helps to explore the connections among multiple variables simultaneously. By employing this advanced approach, we try to provide a more comprehensive understanding of the phenomenon under investigation. Therefore, our study's methodology goes beyond the standard approaches typically used in this issue. Furthermore, by investigating the impact of the shocks of water stress levels and renewable energy consumption on milk production, we can obtain more information from the study results that can contribute to the goals of sustainable development and offer valuable guidance to researchers and policymakers in this issue.

3. Data and Methods

This section briefly presents the types of data used and the econometric approaches applied for data analysis in our study.

3.1. Data

Annual time series data were collected covering the period from 2000 to 2021 in Saudi Arabia. The selected variables used in the study are total milk production (TMP) in tons (t), water stress levels in agriculture (WSA) in (%), water stress levels in industry (WSI) in (%), and renewable energy consumption (REC) in terajoules (TJ). The limited study period was constrained by the availability of time series data regarding water stress. Data representing water stress levels and milk production were obtained from the FAO database [45], while data concerning REC were obtained from the World Bank [46]. We chose these specific variables for investigation because milk is a vital source of nutrition, particularly for children, and ensuring a sufficient and sustainable milk supply is vital for food security. Analyzing water stress in Saudi Arabia helps to assess the water supply for dairy activities, promote efficient irrigation practices, and ensure the long-term sustainability of milk production. Furthermore, we chose the renewable energy consumption variable because it plays an increasingly significant role in Saudi Arabia's energy sector as part of its efforts to diversify its economy, reduce dependence on fossil fuels, and contribute to establishing a sustainable and environmentally friendly energy sector.

The descriptive statistics, pairwise correlations, and normality tests for the selected variables are presented in Table 1. Strong positive significant connections were observed between total milk production and water stress levels in industry (correlation coefficient, $r = 0.93$ at the 1% level) and between water stress levels in agriculture and industries ($r = 0.51$ at the 1% level). Hypothetically, this might be due to government policies or incentives applied to support and promote dairy farming in Saudi Arabia. The null hypotheses for skewness and kurtosis tests were rejected at the 5% significant level, while the Shapiro–Wilk normality test null hypothesis (H_0) was rejected at the 10% and 5% significant levels specifically for total milk production and renewable energy consumption, respectively, which is evidence of leptokurtic distributions, implying that these variables are not normally distributed. Consequently, they were transformed into natural logarithm forms to reduce positive skewness in the data.

Table 1. Matrix correlation and descriptive statistics.

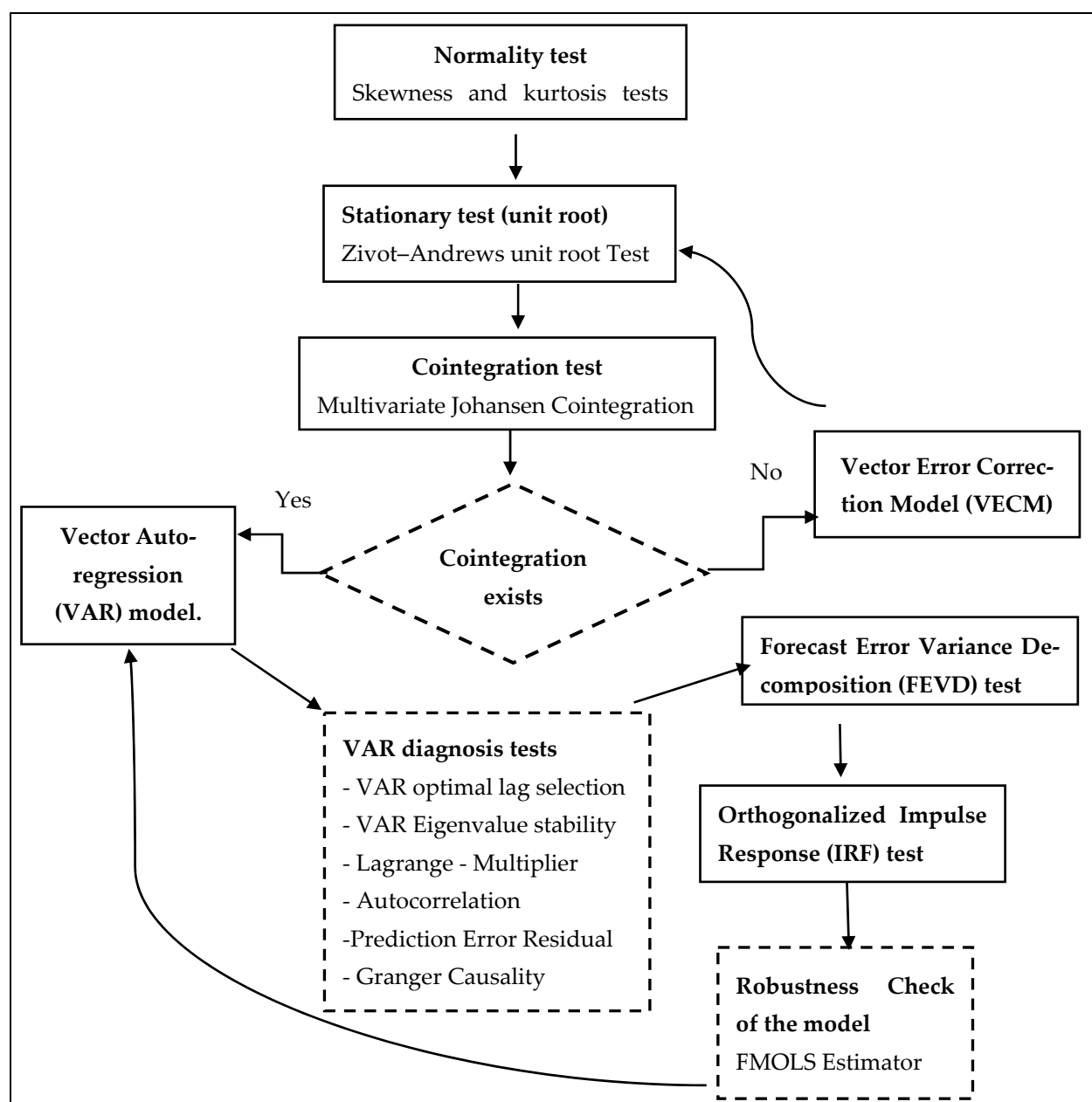
1. Correlation Matrix				
Variable	TMP	WSA	WSI	REC
TMP	1.00			
WSA	0.29 (0.19)	1.00		
WSI	0.93 *** (0.00)	0.51 *** (0.01)	1.00	
REC	−0.06 (0.78)	−0.006 (0.98)	−0.15 (0.51)	1.00
2. Descriptive Statistics				
Mean	1,873,405.00	776.70	33.51	232.94
Std. dev.	645,574.60	27.99	12.08	35.73
Min	952,500.00	719.17	11.35	181.71
Max	2,916,104.00	830.15	52.49	285.33
3. Normality Tests:				
A. Skewness and kurtosis tests				
Statistics test	Pr (skewness)	Pr (kurtosis)	Joint test	
			chi2(2)	Prob > chi2
TMP	0.864	0.017	5.71	0.058
WSA	0.648	0.788	0.28	0.869

Table 1. *Cont.*

WSI	0.890	0.695	0.17	0.917
REC	0.611	0.035	4.73	0.094
B. Shapiro–Wilk W test				
Statistics test	W	V	Z	Prob > z
TMP	0.922	1.982	1.387	0.083
WSA	0.971	0.732	−0.632	0.736
WSI	0.943	1.441	0.740	0.229
REC	0.908	2.340	1.724	0.042

Note: W = Shapiro–Wilk test statistic (W); V = related to the critical values used to determine the statistical significance of W. *** refers to the level of significance at 1%. Source: authors' calculation (2024).

The steps and models designed for the study are demonstrated in Figure 2.

**Figure 2.** Flowchart of the methodical framework. Source: authors' design (2024).

3.2. Approaches

3.2.1. Unit Root and Cointegration Tests

This study employed a vector autoregressive (VAR) technique and its environment, which are generally used for forecasting systems of interconnected time series in the macroeconomic literature [47].

Furthermore, the examination of the dynamic impact of random disturbances on a system of variables can be approached through alternative methodologies, expanding the existing literature beyond the traditional multiple simultaneous equation models proposed by [48] as referenced in [49]. This approach allows for a more comprehensive analysis of the system's behavior and provides valuable insights into the effects of random disturbances on the variables under examination.

An empirical analysis was conducted to perform unit root and cointegration tests before investigating the VAR model. As most time series have a structural break, the unit root test established by [50], abbreviated as ZA, is used. This unit root test adopts the sequential Augmented Dickey–Fuller (ADF) [51] test to find the break corresponding to the alternative three models. Model (A) allows for a change in the intercept of the series, Model (B) allows for a change in the trend of a series, while Model (C) allows for changes in both intercept and trend. Most researchers [52,53], use Model A and/or Model C. We approved Model C for the investigation of unit roots in this study, as it permits a breakpoint in trend and intercept.

We applied the ZA test for analyzing the unit root for milk production, water, and renewable energy consumption nexuses due to its ability to account for potential structural breaks and accurately evaluate the stationarity properties of the data and because this is crucial to conducting further analysis, such as modeling, forecasting, or policy evaluation, in these domains.

Moreover, this study examined the Johanson–Juselius cointegration technique [54] to explore the long-run equilibrium connection among the four selected variables (TMP, WSA, WSI, and REC) and to detect the number of cointegrating vectors. The justification for adopting this test is that it helps to identify whether milk production, water stress level, and renewable energy consumption are cointegrated and provides insights into their long-term interdependencies, allowing for better modeling, forecasting, and policy decisions regarding these domains. Also, the proposal for adopting the VAR model relies on the cointegration test results. Johansen suggests two rank test statistics (Trace and Max-Eigen) to check H_0 , which the numbers of characteristic roots suggested to be insignificantly different from unity, and these ranks take the formula as follows:

$$\gamma_{trace}(r) = -N \sum_{i=r+1}^n \ln(1 - \hat{\gamma}_i) \quad (1)$$

$$\gamma_{max-eigne}(r, r+1) = -N \ln(1 - \hat{\gamma}_{i+1}) \quad (2)$$

where $\hat{\gamma}_i$ is the estimated characteristic, N denotes the number of observations, γ_{trace} tests the H_0 of $r = 0$ against H_1 of $r > 0$, and $\gamma_{max-eigne}$ tests the H_0 of $r = 0$ against H_1 of $r = 1$.

3.2.2. VAR Model

In a VAR system, all variables are considered endogenous, but certain identifying restrictions can be imposed based on theoretical frameworks or statistical methods to understand the effects of exogenous shocks on the system [55]. The VAR model can be identified when assuming K times series variables $Y_t = (Y_{1t}, \dots, Y_{Kt})$, as follows:

$$Y_t = \alpha + A_1 y_{t-1} + \dots + A_\rho y_{t-\rho} + u_t = \alpha + A Y_{t-1}^{\rho} + u_t \quad (3)$$

where Y_t is the $K \times 1$ vector of endogenous variables, in our study represented by TMP, WSA, WSI, and REC, which are being forecasted. Let $A_1, A_2, \dots$ represent the coefficient matrices for the polynomial matrices of lag length ρ , denoted by $(K \times n)$. The vector (u_t) ,

of size $(n \times 1)$, represents the white-noise error terms, including the reduced form residuals, which typically exhibit non-zero correlations; α denotes a constant term, while t signifies the number of samples. The VAR model incorporates lagged values of all variables as endogenous; consequently, a separate equation is created for each variable. Thus, the VAR equations used in this study take the following forms:

$$\text{LnTMP}_t = \alpha_1 + A_{11}\text{LnTMP}_{t-1,3} + A_{12}\text{LnWSA}_{t-1,3} + A_{13}\text{LnWSI}_{t-1,3} + A_{14}\text{LnREC}_{t-1,3} + u_{t,1} \quad (4)$$

$$\text{LnWSA}_t = \alpha_2 + A_{21}\text{LnTMP}_{t-1,3} + A_{22}\text{LnWSA}_{t-1,3} + A_{23}\text{LnWSI}_{t-1,3} + A_{24}\text{LnREC}_{t-1,3} + u_{t,2} \quad (5)$$

$$\text{LnWSI}_t = \alpha_3 + A_{31}\text{LnTMP}_{t-1,3} + A_{32}\text{LnWSA}_{t-1,3} + A_{33}\text{LnWSI}_{t-1,3} + A_{34}\text{LnREC}_{t-1,3} + u_{t,3} \quad (6)$$

$$\text{LnREC}_t = \alpha_4 + A_{41}\text{LnTMP}_{t-1,3} + A_{42}\text{LnWSA}_{t-1,3} + A_{43}\text{LnWSI}_{t-1,3} + A_{44}\text{LnREC}_{t-1,3} + u_{t,4} \quad (7)$$

To determine the appropriate lag length model, we utilized lag length selection criteria, including the Akaike Information Criterion (AIC), Schwarz–Bayesian (SBIC) criterion, Hannan–Quinn criterion (HQIC), Likelihood Ratio, sequential modified (LR) criteria, and final prediction error (FPE) based on [56]. We approved the model that exhibited the lowest Akaike Information Criterion (AIC) or Schwarz–Bayesian (SBIC) value, specifying the best fit. Based on Equation (3), to estimate the VAR order, ρ , we employed equation-wise ordinary least squares (OLS) with the inclusion of three lags in this study.

The justification for applying the VAR model and its environment is that it enables the simultaneous analysis of multiple variables, making it suitable for examining the interactions between milk production, water stress levels, and renewable energy consumption, which can all influence each other, and helping to disentangle the causal relationships among variables. By investigating VAR, we can evaluate the impact of external shocks and policy interventions, such as variations in renewable energy policies and water stress levels, on milk production. Additionally, we can quantify and understand the magnitude of these shocks and help forecasting supporting policy formulation and decision making by providing a clear and robust picture of our findings through checking the Granger causality, IRF, and FEVD tests.

3.2.3. Granger Causality Test

The Granger causality test is a commonly utilized method in studies considering energy and water investigations to establish causal connections between variables [57,58]. The current study applies the Granger causality test recommended by [59] for detecting the causal relationship among the selected variables, with a VAR framework. The pairwise Granger causality model can be written as

$$\text{LnTMP}_t = \phi_1 + \sum_{j=1}^n \phi_{1j}\text{LnWSL}_{t-1} + \sum_{j=1}^n \phi_{2j}\text{LnTMP}_{t-j} + \omega_{1t} \quad (8)$$

$$\text{LnTMP}_t = \alpha_1 + \sum_{j=1}^m \alpha_{1j}\text{LnREC}_{t-1} + \sum_{j=1}^m \alpha_{2j}\text{LnTMP}_{t-j} + \omega_{2t} \quad (9)$$

$$\text{LnWSL}_t = \beta_1 + \sum_{j=1}^p \beta_{1j}\text{LnTMP}_{t-1} + \sum_{j=1}^p \beta_{1j}\text{LnWSL}_{t-j} + \epsilon_{3t} \quad (10)$$

$$\text{LnWSL}_t = \delta_1 + \sum_{j=1}^q \delta_{1j}\text{LnREC}_{t-1} + \sum_{j=1}^q \delta_{2j}\text{LnWSL}_{t-j} + \epsilon_{4t} \quad (11)$$

$$\text{LnREC}_t = \gamma_1 + \sum_{j=1}^r \gamma_{1j}\text{LnTMP}_{t-1} + \sum_{j=1}^r \gamma_{2j}\text{LnREC}_{t-j} + \epsilon_{5t} \quad (12)$$

$$\text{LnREC}_t = \phi_1 + \sum_{j=1}^s \phi_{1j}\text{LnWSL}_{t-1} + \sum_{j=1}^s \phi_{2j}\text{LnREC}_{t-j} + \epsilon_{6t} \quad (13)$$

where WSL are the water stress levels, which could be in agriculture (WAS) or in industry (WSI); $\emptyset$, α , δ , γ , and φ are intercepts of the equations, and their corresponding j represents the coefficients of the equations; n , m , p , q , r , and g represent the lags order; and ω_{it} ($i = 1$ to 6) are uncorrelated error terms for the equations. The justification for examining the Granger causality test is that it assists in determining the direction and strength of influence among the selected variables, offering valuable insights for policy decisions and efficient allocation of resources in the fields of milk production, water stress levels, renewable energy consumption, and milk production.

3.2.4. Impulse Response Functions and Variance Decompositions

The estimation of IRFs and FEVDs is a crucial aspect of the VAR analysis methodology. To examine the progress of variables' shocks and how the TMP, WSA, WSI, and REC variables affect other variables, impulse response function (IRF) and variance decomposition (FEVD) analyses within the context of VAR were conducted.

Impulse response functions provide insights into whether innovations have positive or negative effects and whether these effects are short-term or long-term in nature [48,60]. Our study applied the orthogonalized impulse response function analysis to the order of the VAR. While impulse response functions (IRFs) trace the effects of a one-standard-deviation shock on present and future values of endogenous variables within the VAR framework, IRFs do not quantify the magnitude of these impacts [61,62].

For assessing the magnitude of the impacts of the selected variables, the FEVD method is utilized. This method determines the percentage contribution of each innovation to the forecast error variance of the dependent variable at a specific time horizon (h -step). We adopted the FEVD method because it allows for a comprehensive analysis of how changes in one variable influence the others, helping in the identification of key drivers of milk production and helping the decision making regarding formulating strategies and policies in milk production. This study also applied the Fully Modified OLS (FMOLS) approach for a robustness check of the models' results.

4. Results and Discussion

4.1. Unit Root and Cointegration Results

The results of the ZA unit root test displayed in Appendix A show that all the selected variables were found to be non-stationary in terms of their levels. In the intercept situation outcomes, the structural break date observed was in 2015 for TMP and WSI; however, significant break dates were identified as 2007 and 2016 for WSA and REC, respectively. Whereas in both trend and intercept models, we noted dissimilar break dates for the selected variables, on the trend side, the breaks were 2009, 2011, 2008, and 2017 for TMP, WSA, WSI, and REC, respectively; on the other hand, the breaks were 2015, 2007, 2018, and 2016 for TMP, WSA, WSI, and REC, respectively, under the trend and intercept conditions (Appendix A).

By recognizing these break dates, we can detect factors driving the trends of the selected variables and facilitate informed decision making regarding resource management and the transition to sustainable energy. For instance, these break dates represent significant changes that impacted milk production, where these changes could have been influenced by shifts in consumer demand or changes in farming systems. Thus, the Zivot–Andrew test suggested non-stationarity with structural breaks in the time series data.

Appendix B presents the results of the Johansen co-integration tests, for the two ranks, Trace and Max-Eigen statistics. The results reveal that the trace statistic value is 50.057, which is less than the 1% of the critical value of 54.681; also, all the p -values are more than 0.05. This indicates that we cannot reject the H_0 of the Johansen cointegration tests, indicating no co-integration among the variables. The same result was obtained from the Max-Eigen test. Ref. [63] proposed that the vector autoregression method is applied when no cointegration exists between two series. This result suggests that variations or fluctuations in milk consumption are not directly related to changes in water stress or

renewable energy consumption, and vice versa, in the long run. Consequently, variations and fluctuations in milk production can indirectly impact the trend of greenhouse gas emissions from the dairy sector, contributing to climate change globally [64]. This is because dairy products are consumed worldwide, and the dairy sector is a major industry in many countries.

The three variables may show unrelated trends and behaviors, and their movements are not bound together in a stable, long-term manner. Therefore, in the absence of cointegration, we would not expect to observe a consistent, predictable connection among milk production, water, and energy consumption over an extended period. Therefore, when investigating these variables together, it would be more appropriate to focus on short-term dynamics or consider alternative modeling approaches that are not based on the hypothesis of a long-term relationship, such as VAR. As a result, the selected variables are not co-integrated, so the application of the VAR model can be justified to be a suitable approach to capture the dynamics and interrelationships among the selected variables. i.e., there is a higher probability that Equation (3) can be employed as a suitable model for making accurate forecasts.

4.2. VAR Results

Table 2 highlights that the VAR results indicate significant coefficients for the selected variables in specific lag periods. However, it is observed that there is a lack of significant interdependence among certain selected variables. This implies that these variables do not exert influence on one another or are not influenced by each other. Thus, there is a negative significant interdependence between LnTMP and LnWSA (-2) (p -value < 0.01), but a positive significant connection between LnTMP and LnWSA (-3) (p -value < 0.01) is also observed. Furthermore, a negative significant relation is noted between LnREC and LnWSA (-1) (p -value < 0.01). Moreover, there is a positive interconnection between LnTMP and WSI at lags 1, 2, and 3, which proves the findings of the correlation analysis in Table 1. The findings are consistent with [65], who indicate that renewable energy significantly decreased milk production. Furthermore, LnWSA and LnWSI increased by 0.137% and 0.363% when LnREC (-3) decreased by one (TJ), respectively. Similarly, LnREC decreased by −3.023 TJ, while LnWSA (-1) increased by one percentage point. This can be clarified by the fact that renewable energy sources, such as solar and wind power, often require less water for their operation compared with traditional energy sources like coal or natural gas. By using renewable energy sources that require less water, the emissions associated with water-intensive energy production and carbon footprint can effectively be reduced. This allows for mitigating the causes of climate change and contributing to its reduction.

Table 2. Vector autoregression estimates.

Predictor Variable	Response Variable: Equations			
	LnTMP: Equation (4)	LnWSA: Equation (5)	LnWSI: Equation (6)	LnREC: Equation (7)
LnTMP (-1)	0.115 [0.142] (0.809)	−0.044 [0.093] (−0.474)	−0.128 [0.108] (−1.185)	−0.062 [0.324] (−0.190)
LnTMP (-2)	0.213 [0.141] (1.509)	0.037 [0.093] (0.396)	0.151 [0.107] (1.407)	−0.212 [0.322] (−0.660)
LnTMP (-3)	0.184 [0.128] (1.440)	−0.179 [0.084] (−2.133) **	0.363 [0.097] (3.740) ***	−0.344 [0.291] (−1.183)
LnWSA (-1)	−0.164 [0.340] (−0.483)	0.567 [0.222] (2.551) ***	0.088 [0.257] (0.343)	−3.023 [0.772] (−3.917) ***
LnWSA (-2)	−1.610 [0.628] (−2.565) ***	−0.042 [0.411] (−0.103)	0.282 [0.476] (0.592)	2.000 [1.427] (1.401)
LnWSA (-3)	1.516 [0.427] (3.549) ***	−0.572 [0.280] (−2.045) **	0.472 [0.324] (1.458)	−0.858 [0.971] (−0.884)

Table 2. Cont.

Predictor Variable	Response Variable: Equations			
	LnTMP: Equation (4)	LnWSA: Equation (5)	LnWSI: Equation (6)	LnREC: Equation (7)
LnWSI (-1)	0.389 [0.157] (2.483) **	0.021 [0.103] (0.208)	0.355 [0.119] (2.986) ***	0.244 [0.356] (0.685)
LnWSI (-2)	0.313 [0.156] (2.012) **	0.100 [0.102] (0.978)	0.048 [0.118] (0.409)	0.018 [0.354] (0.051)
LnWSI (-3)	−0.185 [0.139] (−1.327)	0.055 [0.091] (0.597)	−0.006 [0.106] (−0.058)	0.237 [0.317] (0.747)
lnREC (-1)	−0.220 [0.122] (−1.812) *	−0.119 [0.080] (−1.498)	0.006 [0.092] (0.067)	1.016 [0.276] (3.677) ***
lnREC (-2)	0.790 [0.178] (4.429) ***	−0.026 [0.117] (−0.223)	−0.573 [0.135] (−4.243) ***	−0.511 [0.405] (−1.261)
lnREC (-3)	−0.460 [0.097] (−4.739) ***	0.137 [0.064] (2.148) **	0.308 [0.074] (4.187) ***	0.380 [0.221] (1.718) *
Constants	6.418 [3.231] (1.986) **	9.085 [2.115] (4.296) ***	−7.527 [2.449] (−3.074) ***	20.301 [7.343] (2.765) ***
PARMS	13	13	13	13
RMSE	0.035	0.023	0.027	0.080
R-sq	0.996	0.790	0.997	0.913
X2	4504.641 ***	71.424 ***	5996.207 ***	200.198 ***

Note: [Std. err.] is in square brackets, and the test statistic (Z) is in parentheses. ***, **, and * refer to levels of significance at 1%, 5%, and 10%, respectively. Source: authors' calculations (2024).

In this context, given the potential negative impacts of reduced renewable energy consumption on water stress levels, economic policymakers can prioritize and incentivize investments in renewable energy projects. Thus, in regions facing water stress, like Saudi Arabia, there may be a greater need for water pumping, desalination, or other energy-intensive methods to meet water demands. However, these practices can have adverse effects, including increased greenhouse gas emissions, which further contribute to the exacerbation of climate change.

As mentioned before, the optimal lag length was selected based on the AIC, SC, and HQ information standards. From Table 3, the minimum statistical value for all the criteria accepted is lag 3 (the lag was taken based on the level that had the most * (asterisks) among all the criteria (indicates the highest criteria preference) as an optimal lag length that satisfies the probability (LR) measure for the reliable VAR model).

Table 3. The optimum lag selection criteria for the estimated VAR model.

Lag	LogL	LR	FPE	AIC	SC	HQ
0	71.393	NA	9.76×10^{-9}	−7.094	−6.895	−7.060
1	145.885	109.778 *	2.17×10^{-11}	−13.251	−12.257	−13.083
2	163.335	18.368	2.47×10^{-11}	−13.404	−11.614	−13.101
3	201.290	23.972	5.97×10^{-12} *	−15.715 *	−13.130 *	−15.277 *

Note: * indicates lag order selected by the criterion based on the following: LR: sequential modified LR test statistic (each test at the 5% level); FPE: final prediction error; AIC: Akaike Information Criterion; SC: Schwarz information criterion; and HQ: Hannan–Quinn information criterion. Source: authors' calculations (2024).

The stability test of the VAR model was examined to ensure the accuracy of the model estimation results. By checking the eigenvalue stability condition, we found that the eigenvalues of the companion matrix lie inside the unit circle and that the real roots are far from one (Table 4). Thus, the VAR model satisfies the stability condition, i.e., the variable's

behavior is predictable and does not exhibit unexpected or uncontrolled behavior over the selected time, i.e., all the eigenvalues (roots) and moduli are less than one, indicating that the four VAR models (Equations (4)–(7)) established between the selected variables are stable and that the various tests based on this model are valid. This result agrees with the study investigating the connection between water and energy [66].

Table 4. Eigenvalue stability condition.

* Root	Modulus
0.987	0.987
0.777 + 0.418 i	0.882
0.777 − 0.418 i	0.882
0.449 + 0.759 i	0.882
0.449 − 0.759 i	0.882
−0.326 + 0.764 i	0.831
−0.326 − 0.764 i	0.831
0.741	0.741
−0.614	0.614
−0.516 + 0.139 i	0.534
−0.516 − 0.139 i	0.053
0.173	0.173

Notes: * Eigenvalues are numbers with both real and imaginary components. The real part of an eigenvalue is represented by the left value, while the imaginary part (i) is represented by the right value. The “+” and “−” signs indicate the sign of the imaginary components of these eigenvalues. Decision: All eigenvalues of the VAR model fall within the unit circle, indicating that the model satisfies the stability condition. Source: authors’ calculations (2024).

The predicted error is estimated as a mean (-1.08×10^{-10}), which is very close to zero. This indicates a high level of accuracy in the model’s predictions. Furthermore, the Lagrange multiplier residual autocorrelations (LM) test was performed, and it is obvious from the findings that the p -value is greater than 5% for the LM test (probability values of the four lag orders estimated as 0.70301, 0.184, 0.431, and 0.269), indicating that we failed to reject H_0 ; hence, there is no autocorrelation in lag order (Table 5).

Table 5. Autocorrelation and prediction error residual test results.

Lagrange Multiplier Test			
Lag	χ^2	$P > \chi^2$	Decision
1	12.583	0.703	H_0 accepted
2	20.865	0.184	H_0 accepted
3	16.322	0.431	H_0 accepted
4	18.978	0.270	H_0 accepted
Predict error, residual	Mean (Std. dev)	[Min] [Max]	
Value	-1.08×10^{-10} (0.039)	[−0.081] [0.079]	Accuracy in the model’s predictions

Note: H_0 : no serial correlation. Source: authors’ calculations (2024).

4.3. Granger Causality Approach Results

The empirical findings of the Granger causality test in Table 6 reveal a bidirectional causal relationship for most of the trials at different percent levels of significance, suggesting that there is a feedback loop among the selected variables. This implies that variations in water stress levels and renewable energy consumption can affect the level of total milk

production, ($Ch^2 = 21.876, 23.235$, and 46.222 for WSA, WSI, and REC, respectively), while changes in total milk production can also influence water stress level and renewable energy ($Ch^2 = 9.6907, 39.863$, and 8.0037 for WSA, WSI, and REC, respectively). This explains that when the water stress level increases, it may lead to a decrease in available water resources for irrigation, which could affect fodder production for dairy cows, which, indirectly, might result in reduced milk production. Conversely, an increase in total milk production may require additional renewable energy resources, for cooling, processing, and other services, which could impact the demand for renewable energy sources. Moreover, the results propose a unidirectional causality relation running from renewable energy to water stress levels in industry. This indicates that continuous renewable energy consumption simultaneously creates a continuous rise in water stress levels in industry. However, the WSI variable does not present Granger causality with respect to REC (p -value = 0.179). Generally, by considering these complex relationships, policymakers, investigators, and stakeholders can work towards sustainable practices that balance the needs of water availability, milk production, and renewable energy demand in the context of a changing climate and evolving agricultural industry. For more confirmation, [67] stated that the concepts of food, water, and energy are increasingly examined together by applying a so-called nexus approach.

Table 6. Granger causality results.

Equation: LnTMP			
Excluded	Chi ²	<i>p</i> -Value	Causality, Direction
LnWSA	21.876	0.000 ***	WSA $\leftrightarrow$ TMP, bidirectional
LnWSI	23.235	0.000 ***	WSI $\leftrightarrow$ TMP, bidirectional
LnREC	46.222	0.000 ***	REC $\leftrightarrow$ TMP, bidirectional
All	121.55	0.000 ***	Causality
Equation: LnWSA			
Excluded	Chi ²	<i>p</i> -value	Causality, direction
LnTMP	9.6907	0.021 **	TMP $\leftrightarrow$ WSA, bidirectional
LnWSI	7.6562	0.054 *	WSI $\leftrightarrow$ WSA, bidirectional
LnREC	17.668	0.001 ***	REC $\leftrightarrow$ WSA, bidirectional
All	24.454	0.004 ***	Causality
Equation: LnWSI			
Excluded	Chi ²	<i>p</i> -value	Causality, direction
LnTMP	39.863	0.000 ***	TMP $\leftrightarrow$ WSI, bidirectional
LnWSA	12.841	0.005 ***	WSA $\leftrightarrow$ WSI, bidirectional
LnREC	89.934	0.000 ***	REC $\rightarrow$ WSI, unidirectional
All	236.05	0.000 ***	Causality
Equation: LnREC			
Excluded	Chi ²	<i>p</i> -value	Causality, direction
LnTMP	8.0037	0.046 *	TMP $\leftrightarrow$ REC, bidirectional
LnWSA	18.66	0.000 ***	WSA $\leftrightarrow$ REC, bidirectional
LnWSI	4.908	0.179	No causality
All	42.218	0.000 ***	Causality

Note: ***, **, and * denote significance at 1%, 5%, and 10%, respectively. Source: authors' calculations (2024). $\leftrightarrow$ It means a bidirectional run from both the left and right variables and $\rightarrow$ means unidirectional run from the left variable.

4.4. Forecast Error Variance Decomposition Results

To effectively illustrate the dynamics of the VAR system, variance decomposition is an appropriate technique. It provides insights into the key drivers, shock broadcast, and forecasting precision. By applying the FEVD technique, we can detect the main drivers of variability of the selected variables, so policymakers can plan targeted interventions or policies to alleviate risks and suggest more informed decisions for milk production, water stress level, and renewable energy demand. Therefore, it is essential to decompose the shocks of an endogenous variable (TMP, WSA, WSI, and REC) by attributing them to specific shocks associated with other variables within the model. In this study, we selected, for example, horizon 2 to denote the short run and horizon 10 to denote the long run. We note that a portion of nearly 85.921% of TMP is significantly attributable to its innovative shock in the short run compared with 37.71% in the long run (Table 7).

Table 7. Variance decomposition results.

Horizons	Variance Decomposition of LnTMP				Variance Decomposition of LnWSA			
	TMP	WSA	WSI	REC	TMP	WSA	WSI	REC
1	100.000	0.000	0.000	0.000	0.701	99.300	0.000	0.000
2	85.921	1.775	1.088	11.216	2.007	89.840	0.169	7.985
3	46.518	2.872	7.245	43.365	8.135	69.488	0.265	22.112
4	40.650	11.481	7.135	40.734	7.462	63.726	0.249	28.563
5	43.004	11.135	6.609	39.252	7.389	63.791	0.247	28.574
6	38.167	23.914	5.481	32.438	7.250	63.010	0.602	29.138
7	38.808	28.140	4.415	28.637	7.340	62.948	0.682	29.031
8	38.095	25.549	3.539	32.817	7.258	62.383	0.675	29.684
9	37.254	25.466	3.330	33.950	7.149	62.122	0.689	30.040
10	37.706	24.075	3.151	35.068	7.383	61.220	0.772	30.624
Horizons	Variance Decomposition of LnWSI				Variance Decomposition of LnREC			
	TMP	WSA	WSI	REC	TMP	WSA	WSI	REC
1	5.542	50.771	43.687	0.000	13.002	26.082	2.276	58.641
2	5.323	52.801	41.859	0.017	17.072	17.862	3.397	61.670
3	18.127	22.781	14.635	44.457	19.546	15.388	3.797	61.269
4	27.541	26.568	6.786	39.104	26.255	12.006	3.490	58.249
5	28.739	25.045	5.011	41.205	26.042	10.279	3.158	60.522
6	32.806	23.162	4.090	39.942	26.191	10.519	3.127	60.163
7	32.211	21.956	3.781	42.051	25.271	14.262	2.972	57.495
8	32.386	22.284	3.820	41.509	25.393	15.165	2.943	56.499
9	32.569	23.126	3.693	40.612	25.721	15.094	2.934	56.252
10	34.268	22.343	3.590	39.799	26.334	15.090	2.904	55.672

Note: Cholesky ordering: LnTMP, LnWSA, LnWSI, and LnREC. Source: authors' calculations (2024).

The contribution of REC to the shocks of TMP varies between 11.22% (in the short run) and 35.01% (in the long run) more than the contributions of WSA (1.77–24.08%) and WSI (1.08–3.15%). Moreover, the contributions of WSA and REC to TMP FEVDs show a growing trend, which suggests that water stress levels in agriculture are becoming a more dominant driver of variations in total milk production in Saudi Arabia. This trend highlights the significance of addressing water stress levels in agriculture and realizing strategies to minimize their influence on agriculture, particularly dairy farming, to ensure sustainable and resilient food production systems. Identifying the significance of water stress levels in

agriculture, specifically in dairy farming, positively influences social well-being by ensuring a stable and secure food supply, contributing to economic stability and growth, enhancing climate change resilience, and contributing to sustainable agricultural practices. When comparing our results with the study performed in Taiwan, which applied the life cycle assessment (LCA) protocol, we find that the study agrees with our findings. Therefore, water is of absolute importance for the production and profitability of raw milk [22].

Likewise, the growing trend in the contributions of REC to milk production innovation indicates a shift towards incorporating more sustainable practices on dairy farms.

TMP, WSI, and REC shocks explain almost 2%, 0.17%, and 8% of the uncertainty in WSA in the short run, respectively. These percentages increase to 7.38%, 0.8%, and 30.62% in the long run, respectively. It also observed that the shocks of WSA largely occur in the short and long run. The innovative shocks of WSI are enhanced in TMP by 1.09% and 3.015% in the short and long run, respectively.

Also, we detected that in the first horizon (1) the FEVDs of TMP and WSA are completely determined by their innovation shocks. The contribution of REC to the WSI shock is small (0.017%) in the short run, whereas REC explains the great extent of FEVDs in the long term in WSI, with a contribution of 39.78%. The contribution of WSI to its innovation shock is large in the short run (41.90%) and shrinks slowly in the long run (3.60%). The contributions of TMP and WSA to the shocks of REC are nearly similar in the short run; however, the variation in REC in the short run and the long run is determined largely by its innovation shocks, 61.67% and 55.67%, respectively.

Finally, we concluded that the variations in TMP, WSA, and REC are largely determined by their shocks through forecasting horizons. Furthermore, the contribution of WSI to WAS and REC variations is minimal in all horizons.

4.5. Impulse Response Function Results

The impulse response functions in the VAR model trace the effect of a shock on the endogenous variables. Figure 3 illustrates the orthogonalized impulse response (OIR) of the selected variables. The orthogonalized impulse response of TMP generated a one-standard-deviation shock on water stress and renewable energy consumption, demonstrating a positive trend; subsequently, this positive effect gradually increased over time, reaching its maximum positive impact in 10 periods of WSA impulse. As a result, milk production increases when both water stress and renewable energy are present. This could suggest that the introduction of renewable energy sources, despite the presence of water stress, positively affects milk production. This result is confirmed by [65], who revealed that milk production was mainly dependent on renewable energy. This also could suggest that water stress motivates or necessitates greater milk production efforts or the use of renewable energy sources in response to water scarcity or limitations. This suggests that when water availability becomes limited due to water stress, policymakers or farmers may implement strategies to increase milk production despite the constraints. This could involve managing water resources more effectively by implementing more efficient irrigation techniques and adopting drought-tolerant fodder. We conclude that water stress shocks are significant in the whole sample according to impulse response functions.

Further, the results in Figure 4 imply that positive residuals in the milk production curve would suggest that the actual milk production is higher than what is predicted by the model. This could imply factors such as improved feed quality, better herd management practices, or favorable environmental conditions that lead to increased milk yields. Positive residuals in milk production are generally desirable, as they indicate higher productivity and efficiency in the industry [68]. Our result agrees with [69], who found a positive residual curve in the lactation period.

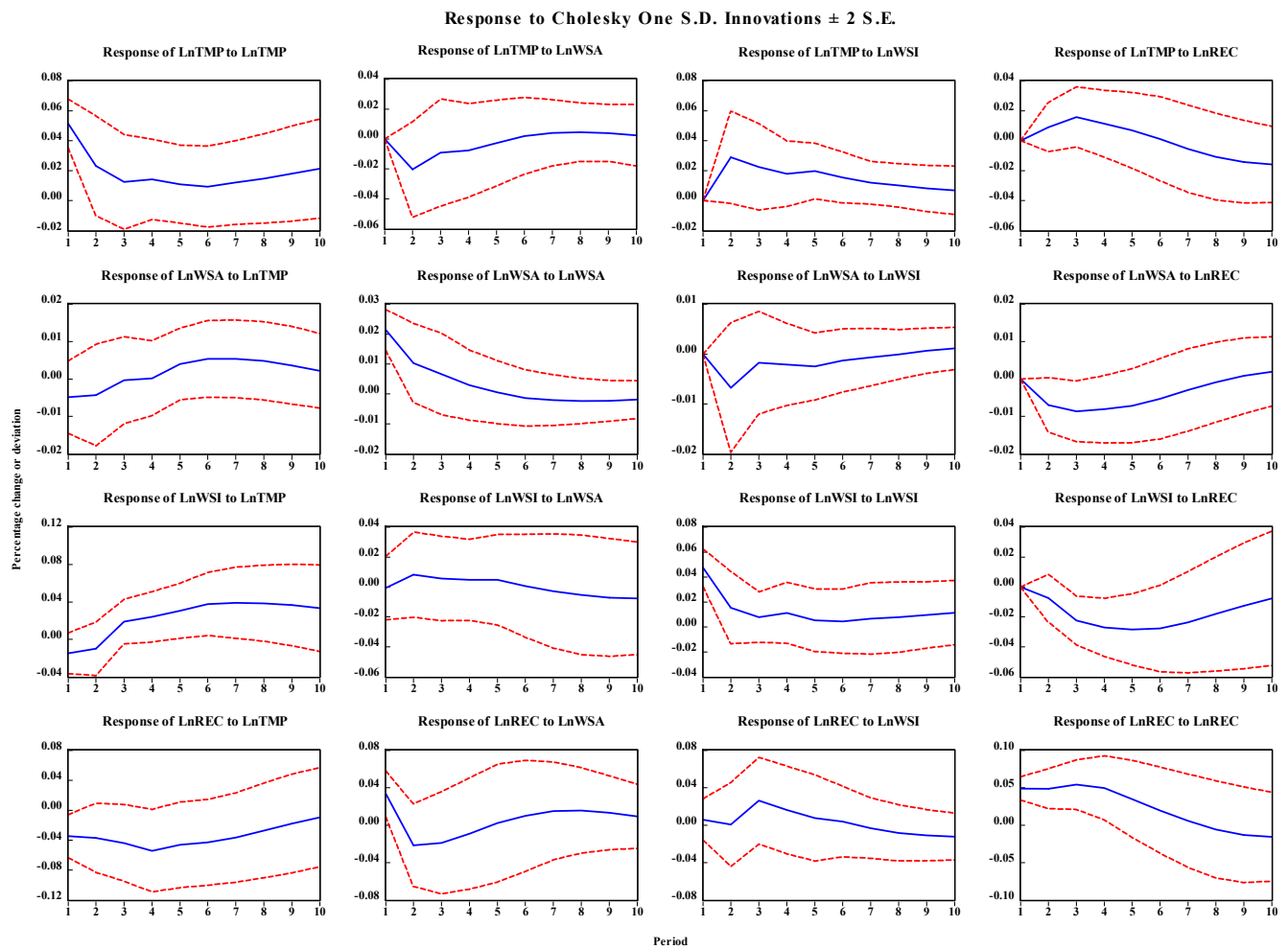


Figure 3. Orthogonalized impulse response. Note: The solid blue lines display the mean responses to a one-standard-deviation shock, while the dotted red lines symbolize the range of ± 2 standard deviations of the responses. The x -axis is the forecast horizon, and the y -axis is the forecasted response of the response variable (TMP, WSA, WSI, or REC) to a unit shock in the corresponding error term.

Negative residuals in water stress levels indicate that the observed water stress is lower than what is predicted. This may suggest effective water management practices, conservation efforts, or improvements in water infrastructure and efficiency. Negative residuals in water stress levels signify a relatively lower level of water scarcity or a successful reduction in water stress.

Positive residuals in REC indicate success in expanding the proportion of renewable energy in the energy mix, while negative residuals may indicate the need for further efforts to promote renewable energy adoption and overcome barriers. This result also agrees with [70], indicating that improving the proportion of renewable and clean energy within the overall energy composition aims to mitigate its environmental impact. Investigating these residual tests provides insights into the efficiency of water management strategies, productivity in the milk industry, and progress in renewable energy adoption. This information can support policymakers, stakeholders, and industry businesses in recognizing areas for development, addressing tasks, and progressing sustainable practices.

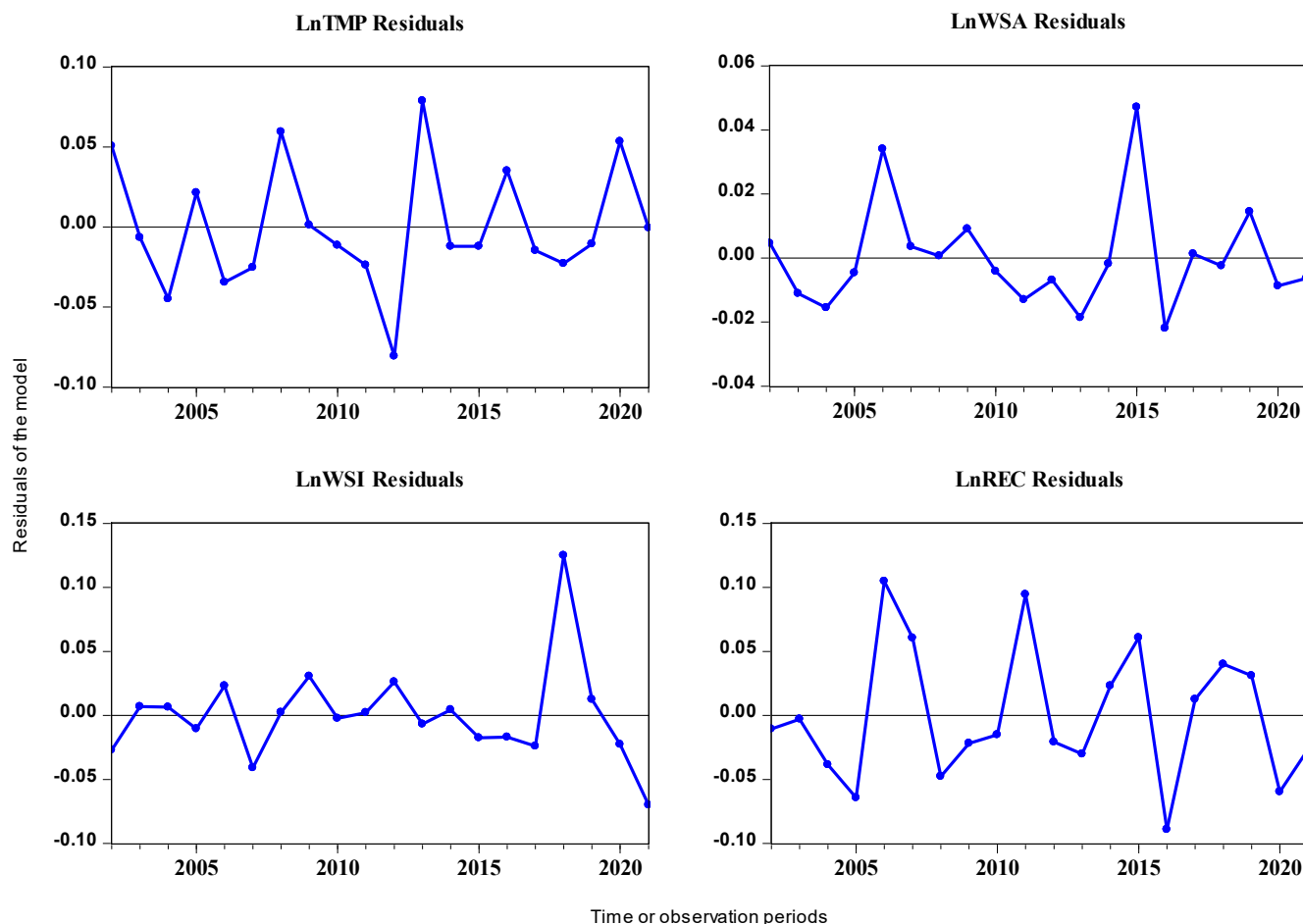


Figure 4. Residual tests. Note: The plot shows residuals for each selected variable obtained from the model. The x -axis represents time, while the y -axis represents the residuals. The residuals are calculated as the differences between observed and predicted values. A solid horizontal blue line at zero is a reference point to detect deviations from expected values. The residuals are randomly scattered around zero, suggesting that the model meets linearity, independence, and constant variance assumptions.

4.6. Robustness Check: FMOLS Estimator Results

Finally, for more robustness of the VAR model, we used the FMOLS estimator to analyze the long-run cointegration of milk production–water stress–renewable energy consumption nexus. While VAR and FMOLS are distinct techniques, they can be related when analyzing cointegrated time series data. We used FMOLS for the robustness check because it estimates the long-run relationships and corrects for endogeneity in VAR models when dealing with nonstationary variables with stable linear combinations. Additionally, the FMOLS method can assist policymakers by providing more accurate estimates of the relationships between milk production, water level stress, and renewable energy consumption. The results of the FMOLS estimator, as shown in Table 8, disclose that water stress in agriculture has significant a negative effect on milk production ($\beta = -2.016$, $p > 0.00$). Similarly, an increase in water stress in industry boosts milk production ($\beta = 0.032$, $p > 0.00$). On the other hand, renewable energy consumption has a minimal impact on milk production ($\beta = 0.002$, $p > 0.00$). A study performed in Pakistan using an artificial neural network (ANN) method indicated that a unit increase in renewable energy significantly reduced milk yield by 0.02 units [65]. Furthermore, the goodness-of-fit of the model is nearly 93% ($R^2 = 0.927$). This value confirms that 93% of the variance in milk production can be accurately predicted based on the water stress level variables. This can be justified by the

fact that dairy farms require an adequate supply of high-quality forage for milk production and the availability of water directly affects the growth and quality of forage crops.

Table 8. FMOLS estimator.

Predictor Variable	Coefficient (β)	Newey–West Std. Error	t-Statistic
LnWSA	−2.016	0.5180	−3.891 ***
LnWSI	0.032	0.002	20.564 ***
LnREC	0.002	0.000	4.994 ***
C	26.194	3.437	7.621 ***
R-squared		0.927	
S.E. of regression		0.102	
S.D. dependent variance		0.349	
Long-run variance		0.005	
Sum squared residues		0.178	

Note: *** denotes significance at 1%. Source: authors' calculations (2024).

5. Conclusions and Policy Implications

The intensive milk industry requires local water supply and energy for various processes. This study discussed the impact of water stress and renewable energy consumption shocks on milk production in Saudi Arabia by using datasets from 2000 to 2021. For the presence of unit roots, we used the Zivot–Andrews test, and cointegration among the variables was tested by employing the recently developed Johanson–Juselius cointegration technique before investigating the VAR model. The VAR diagnosis tests, Granger causality, FEVD, and IRF tests, as well as the FMOLS estimator for more robustness checks of the VAR model, were applied.

The findings show that significant structural breaks and single-date breaks were marked in all studied variables. No cointegration was observed among the selected variables, allowing for the application of the VAR model. The VAR outcome discloses the presence of a negative significant interdependence between total milk production and water stress levels in agriculture. Furthermore, the results revealed a positive interconnection between total milk production and water stress levels in industry, with the acceptance of lag 3 as an optimal lag period that satisfies the reliability of the VAR model with no autocorrelation in the lag order.

A significant bidirectional causality relationship for most of the trials was observed. The existence of this causal relationship between the selected variables may have significant policy implications regarding water stress and renewable energy consumption and hence sustainable development in the milk industry in Saudi Arabia. The variation in total milk production, water stress levels in agriculture, and water stress levels in industry were largely determined by their shocks through forecasting horizons, with the acceptance of lag 3 as an optimal lag period that satisfies the reliability of the VAR model with no autocorrelation in the lag order. The empirical evidence of the IRFs implies that milk production increases when both water stress and renewable energy are present. We concluded that water stress in agriculture has a significant negative effect on milk production, while the increase in water stress in industry and renewable energy consumption boosts milk production. These findings contribute to global knowledge by emphasizing the importance of sustainable water management practices and the adoption of renewable energy sources for efficient milk production.

Implementing water pricing mechanisms and regulations can incentivize responsible water use and discourage wasteful practices. In the context of setting the agricultural policy for boosting milk production, Saudi Arabia must implement additional initiatives to increase the number of dairy farms and improve the productivity of existing farms. These

initiatives must be focused on refining animal health, genetics, and nutrition to improve milk quantities and quality.

We concluded that VAR and its diagnostic results provide a comprehensive understanding of the water–energy–milk nexus, presenting fact-based perspectives and helping decision making for adopting policies that can yield multiple benefits, including sustainable water resource management, reduced environmental impact, technological advancements, and positive socioeconomic outcomes for the dairy industry.

Furthermore, establishing a clear regulatory agenda that stimulates private sector participation in water conservation and renewable energy projects contributes to sustainable development goals. It is critical to consider sustainable and climate-friendly practices in dairy farming to reduce the environmental impact and mitigate climate change globally. Therefore, the political aspects of this study include the essentials for controlling action, global knowledge sharing, the promotion of sustainable practices, and international collaboration to address water management and conservation and renewable energy in the context of dairy farming and its global impact on climate change.

The same study can be performed in countries other than Saudi Arabia, specifically countries that experience water stress or scarcity. The specific requirements to perform this study may vary depending on the local context. However, some common factors to be considered will include the availability of data (for understanding the interconnections between water stress level, milk production, and renewable energy consumption); the assessment of the water resources in the country; milk production analysis; the evaluation of the energy mix of the country; and finally, the examination of the existing policies, regulations, and incentives related to water management, milk production, and renewable energy.

Author Contributions: Conceptualization, M.A.-M. and R.M.E.; methodology, R.M.E. and M.A.-M.; software, R.M.E. and M.A.-M.; validation, R.M.E., M.A.-M. and F.A., formal analysis, R.M.E. and M.A.-M.; investigation, R.M.E., M.A.-M. and F.A., resources, R.M.E.; data curation, R.M.E.; writing—original draft preparation, R.M.E., F.A. and M.A.-M.; writing—review and editing, R.M.E., F.A. and M.A.-M.; visualization, R.M.E., M.A.-M. and F.A.; funding acquisition, R.M.E. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the Deanship of Scientific Research, King Faisal University, Al-Ahsa, Saudi Arabia, through financial support under Ambitious Researcher Track, Grant 6171.

Data Availability Statement: The datasets created and/or analyzed in the existing study are available from the FAO repository FAOSTAT and the World Bank platform Sustainable Energy for All | DataBank ([worldbank.org](https://data.worldbank.org/)).

Conflicts of Interest: The authors declare no conflicts of interest.

Appendix A

Table A1. Zivot–Andrews Unit Root Test Result.

Variable	Model (A) Intercept *		Model (B) Trend **		Model (C) Trend and Intercept ***	
	t-Stat	SBD	t-Stat	SBD	t-Stat	SBD
LnTMP	−5.599	2015	−5.018	2009	−5.741	2015
lnWSA	−6.035	2007	−5.048	2011	−6.034	2007
LnWSI	−4.625	2015	4.282	2008	−5.450	2018
LnREC	−5.631	2016	−4.658	2017	−7.947	2016

Notes: (1) * The critical values are −5.34, −4.80, and −4.58 at the 1%, 5%, and 10% levels of significance, respectively. (2) ** The critical values are −4.93, −4.42, and −4.11 at the 1%, 5%, and 10% levels of significance, respectively. (3) *** The critical values are −5.57, −5.08, and −4.82 at the 1%, 5%, and 10% levels of significance, respectively. Source: authors' calculations (2024).

Appendix B

Table A2. Multivariate Johansen Cointegration Results.

Trace Statistics Test							Conclusion
Hypothesized No. of CE(s)	Params	LL	Eigenvalue	Trace Statistic	Critical Value (1%)	p-Value	
$r = 0$	20	142.929		50.057	54.682	0.031	H_0 accepted
$r \geq 1$	27	155.791	0.7237	24.333	35.458	0.187	H_0 accepted
$r \geq 2$	32	164.002	0.5601	7.909	19.937	0.475	H_0 accepted
$r \geq 3$	35	167.957	0.309	0.490	6.635	0.484	H_0 accepted
$r \geq 4$	36	167.957	0.024217				
Maximum Eigenvalue test							Conclusion
Hypothesized No. of CE(s)	Params	LL	Eigenvalue	Max-Eigen statistic	Critical Value (1%)	p-Value	
$r \geq 0$	20	142.929		25.724	32.715	0.085	H_0 accepted
$r \geq 1$	27	155.791	0.724	16.424	25.861	0.201	H_0 accepted
$r \geq 2$	32	164.003	0.560	7.419	18.520	0.441	H_0 accepted
$r \geq 3$	35	167.957	0.310	0.490	6.635	0.484	H_0 accepted
$r \geq 4$	36	167.957	0.024				
Information criteria							Conclusion
Hypothesized No. of CE(s)	Params	LL	Eigenvalue	SBIC	HQIC	AIC	
$r \geq 0$	20	142.929		−11.297	−12.098	−12.293	Not selected
$r \geq 1$	27	155.791	0.724	11.535	−12.617	−12.879	Not selected
$r \geq 2$	32	164.003	0.560	−11.607 *	−12.889	−13.200	Selected
$r \geq 3$	35	167.957	0.310	−11.529	−12.931 *	13.271	Selected
$r \geq 4$	36	167.957	0.024				

Note: Trace test and Max-Eigen test indicate no cointegration at the 0.01 level; * denotes selected rank (r). Source: authors' calculations (2024).

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